

Jesse Knight

Research Associate, HIV Modelling
MRC Centre for Global Infectious Disease Analysis
Imperial College London

2026.05 📅
jesse.x.knight@protonmail.com ✉️
+1.226.821.2066 📞
jessexknight 🌐

Education

- 2019.09 – 2023.08 **PhD**, Institute of Medical Science, University of Toronto
- Supervisor: **Sharmistha Mishra**
 - Thesis: **Reexamining assumptions in compartmental models of HIV transmission**
- 2015.09 – 2017.12 **MASc**, Engineering Systems & Computing, University of Guelph
- Supervisors: **Graham Taylor & April Khademi**
 - Thesis: **Voxel-wise image analysis for white matter hyperintensity segmentation**
- 2011.09 – 2015.04 **BEng**, Biomedical, University of Guelph
- Mentor: **Bob Dony**
 - Foci: medical image & signal processing, computational modelling, engineering design

Experience

RESEARCH

- 2024.04 – 2026.06 **Research Associate, HIV Modelling, MRC GIDA**, Imperial College London
- Investigate dynamics of structural factors in HIV using individual-based modelling (R)
 - Support systematic reviews of structural factors among sexual and gender minorities
 - Support teaching, mentorship, and marking within the department
- 2023.09 – 2024.03 **Research Data Science Specialist: Mathematical Modeler, Mishra Lab**, Unity Health Toronto
- 2020.06 – 2023.08 **Data Science Specialist: Mathematical Modeler** (part time)
- 2018.04 – 2019.09 **Data Science Specialist: Mathematical Modeler** (full time)
- Co-developed research questions with academic, public health, & community collaborators
 - Led & supported transmission model design & implementation (R, Python, Matlab)
 - Supported timely epidemic response with model-based evidence (HIV, mpox, covid-19)
 - Analyzed epidemiologic data for model inputs & independent research questions (R)
 - Presented & provided feedback in seminars & meetings about ongoing work
 - Led & co-authored abstracts, presentations, manuscripts, & grants
- 2015.06 – 2015.07 **Technician Assistant, Summer Institute**, Engineering World Health, Rwanda
- Worked with local technicians at Kibuye Hospital to repair medical equipment
 - Pitched & supported engineering design projects at Guelph upon return
- 2014.05 – 2014.08 **Research Assistant, Dr. Aviv**, Dept. Medical Imaging, Sunnybrook Research Institute
- Project: Mapping the impact of collateral circulation in acute ischemic stroke
 - Developed reproducible pipelines & tools for brain MRI analysis (Matlab)
- 2013.05 – 2013.08 **Research Assistant, Dr. Eberl**, Biophysics Interdept. Group, University of Guelph
- Project: A mass-balance model of the human anaerobic colon
 - Designed & ran experiments on model sensitivity to numerical methods (Matlab)

MEMBERSHIPS

- 2025.09 – present **Member**, [Canadian Society for Epidemiology and Biostatistics](#)
2021.01 – 2024.03 **Member**, [Data Sciences Institute, University of Toronto](#)

SERVICE & EXTRACURRICULAR

- 2024.07 – present **Reviewer**, [CIHR Doctoral Research Awards Committee](#) (18 applications)
2024.09 – present **Mentor**, [Imperial Activate Mentoring Program](#) (3 mentees)
2020.11 – present **Maintainer**, [ut-thesis](#): official $\LaTeX$ thesis document class for University of Toronto
 - Rewrote from scratch to refine scope & add documentation; provide user support

2019.11 – 2024.09 **Typesetting**, [Noēsis](#): U of T Undergraduate Journal of Philosophy (5 volumes)
2020.09 – 2023.08 **Mentor**, [UofT IMS Mentorship Program](#) (4 mentees)
2020.09 – 2022.02 **Reviewer**, [SRSA Internal Peer Review Program](#) (3 reviews)
2019.07 – 2021.12 **Website** (2019.07–) & **Exec Producer** (2020.07–), [Raw Talk Podcast](#) (1–10 h/wk)
 - Built website from scratch; co-managed team of 30+ to produce 30–90 min episodes

2013.08 – 2016.12 **Engineering Peer Helper**, [Guelph Engineering Peer Helpers](#) (0–5 h/wk)
 - Led problem-solving workshops & exam-review sessions for 1–100 students

CONTRACT WORK

- 2020.01 – 2026.01 **Invoice Automation**, [The Biglieri Group](#) (Python)
2019.11 – 2024.09 **Typesetting**, [Noēsis](#): U of T Undergraduate Journal of Philosophy, 5 volumes ($\LaTeX$)
2021.03 – 2021.06 **Website**, [IMS Magazine](#) (Web)

MISCELLANEOUS PROJECTS

- 2025.03 [Foodbook](#), a personal recipe website; no ads! (Web)
2019.07 [Atom Filter Lines](#), a package for the Atom editor (RIP); 3000+ downloads (Web)
2017.06 [WMH Papers Table](#), an interactive table of research papers (Web)

Teaching & Supervision

TEACHING

- 2020.03 – 2023.08 **Workshop Lead & $\LaTeX$ Support**, [Map & Data Library, University of Toronto](#)
Intro to $\LaTeX$: 25 total workshops (1.5h each)
Intermediate $\LaTeX$: 15 total workshops (1.5h each)
2015.09 – 2017.12 **Teaching Assistant**, [School of Engineering, University of Guelph](#)
ENGG 4040 Medical Imaging Modalities: 2017.09–12, 2015.09–12
ENGG 4660 Medical Image Processing: 2017.01–04, 2016.01–04
ENGG 3410 Systems and Control Theory: 2017.01–04
ENGG 3390 Signal Processing: 2016.09–12
ENGG 4060 Biomedical Signal Processing: 2016.01–04
 - Developed content & led weekly labs & tutorials for 5–80 students
 - Marked 500+ student lab reports; revised lab manuals for 3 courses
 - Gave 3 full lectures during professor absences

SUPERVISION

- 2024.04 – 2025.08 **Co-Supervisor**, School of Public Health, Imperial College London
1 Undergraduate (course project): 2024.04–05
4 MPH (summer project): 2025.05–08, 2026.05–08
1 MSc (summer thesis): 2026.05–08
- 2022.05 – 2023.08 **Co-Supervisor**, Unity Health Toronto
1 Undergraduate (summer student): 2022.05–08, 2023.05–08

Funding & Awards

GRANTS

- 2025.04 – 2027.03 **Co-Investigator**, NPI: Sharmistha Mishra \$139,736
Team Grant
Canadian Immunization Research Network
Examining the role of herd immunity when estimating mpox vaccine effectiveness with observational data during an evolving outbreak: a simulation study
- 2023.03 – 2025.02 **Co-Investigator**, NPI: Sharmistha Mishra \$500,000
Team Grant: Mpox and Zoonotic Threats
Canadian Institutes of Health Research
Optimizing mpox surveillance strategies and preventing epidemic resurgence: a three-province mathematical modeling study
- 2024.02 – 2025.01 **Co-Investigator**, NPI: Sharmistha Mishra \$100,000
Catalyst Grant
Data Sciences Institute, University of Toronto
Examining biases due to confounders and colliders in observational health data using individual-based simulation models
- 2023.03 – 2023.09 **Co-Applicant**, NPI: Darrell H Tan \$2,000,000
Team Grant: Monkeypox Rapid Research Response
Canadian Institutes of Health Research
Canada-Africa Monkeypox Partnership (CAMP): Characterizing transmission dynamics and evaluating medical countermeasures to inform the clinical and public health response to mpox

FELLOWSHIPS & SCHOLARSHIPS

- 2024.07 – 2026.06 **CIHR Fellowship** \$145,833
Canadian Institutes of Health Research
Modelling the potential impact of addressing structural factors on transmission of HIV and elimination of AIDS among men who have sex with men in Sub-Saharan Africa
• Ranked 1/135 in Selection Committee for Post-PhD Fellowships
- 2022.09 – 2023.08 **Queen Elizabeth II Graduate Scholarship in Science & Technology** \$15,000
Ontario Ministry of Colleges and Universities
Representing female sex workers in transmission models of HIV in Sub-Saharan Africa
- 2019.09 – 2022.08 **Canada Graduate Scholarship – Doctoral** \$105,000
Natural Sciences and Engineering Research Council of Canada
Impacts of heterogeneity on deterministic models of HIV epidemics in Southern Africa
• Ranked 2/110 in Selection Committee for Mathematical Sciences
- 2016.09 – 2017.08 **Ontario Graduate Scholarship – Master's** \$15,000
Ontario Ministry of Colleges and Universities
FLAIR MRI: Template development and improved automated lesion segmentation

2015.09 – 2016.08	Canada Graduate Scholarship – Master’s Natural Sciences and Engineering Research Council of Canada <i>FLAIR MRI: Template development and improved automated lesion segmentation</i>	\$17,500
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AWARDS

2023	Student Leadership Award , University of Toronto <i>Outstanding student leadership, service, & commitment to the university</i>	
2022	Li Ka Shing Excellence Award , Unity Health Toronto <i>Embodies the values & advances the mission of UHT through research & leadership</i>	\$500
2022	Open Fellowship Award , Institute of Medical Science, University of Toronto <i>Academic excellence & progress towards degree completion</i>	\$5000
2022	Stephen Pauker Lusted Award , SMDM 44th North American Meeting <i>Outstanding presentation in quantitative methods & theoretical developments</i>	\$250
2019	Entrance Award , Institute of Medical Science, University of Toronto	\$5000
2017	Engineering Teaching Assistant of the Year , University of Guelph	
2015, 2016	Graduate Dean’s Scholarship , College of Phys & Eng Science, University of Guelph <i>Outstanding academic & research performance</i>	\$1125
2015, 2016	Engineering Peer Helper of the Year , University of Guelph	
2015	E.B. MacNaughton Convocation Prize , University of Guelph <i>College nominee for the W.C. Winegard Medal: top overall convocation award</i>	
2015	Helen Grace Tucker Design Prize , University of Guelph <i>Best performance in engineering design courses, years 2–4</i>	\$500
2015	Association of the Professional Engineers Medal , University of Guelph <i>Highest average all coursework in graduating year</i>	
2015	Society of Excellence , College of Phys & Eng Science, University of Guelph <i>Academic achievement & contributions to the U of G community</i>	
2013, 2014	Dean’s Scholarship , University of Guelph	\$1000

Publications

JOURNAL ARTICLES

1. **Jesse Knight**, Huiting Ma, Bheki Sithole, Lungile Khumalo, et al. “Evaluating the impact of achieving cascade equality in Eswatini: a modeling study on the prevention impacts of antiretroviral therapy”. *AIDS* 40.3 (Mar. 2026), pp. 362–371. doi.org/10.1097/qad.0000000000004400.
2. James Stannah, **Jesse Knight**, Theo Sandfort, Christian Laurent, et al. “The effect of sexual and gender minority violence on depression, hazardous drinking, condom use, and HIV acquisition: an individual participant data meta-analysis of the CohMSM, HPTN 075, and Anza Mapema cohort studies in Africa”. *AIDS and Behavior* 29 (July 2025), pp. 3557–3572. doi.org/10.1007/s10461-025-04799-4.
3. Julia Michalow, Lauren Hall, Jane Rowley, Rebecca L Anderson, et al. “Prevalence of chlamydia, gonorrhoea, and trichomoniasis among male and female general populations in sub-Saharan Africa from 2000-2024: A systematic review and meta-regression analysis”. *eClinicalMedicine* 83 (May 2025), p. 103210. doi.org/10.1016/j.eclinm.2025.103210.
4. Fanyu Xiu, Carla Doyle, Jorge Luis Flores Anato, **Jesse Knight**, et al. “Impact of interventions on mpox transmission during the 2022 outbreak in Canada: a mathematical modeling study of three different cities”. *International Journal of Infectious Diseases (IJID)* 153 (Apr. 2025), p. 107792. doi.org/10.1016/j.ijid.2025.107792.
5. Katherine Rucinski, **Jesse Knight**, Kalai Willis, Linwei Wang, et al. “Challenges and opportunities in big data science to address health inequities and focus the HIV response”. *Current HIV/AIDS Reports* 21 (June 2024), pp. 208–219. doi.org/10.1007/s11904-024-00702-3.

6. Fanyu Xiu, Jorge Luis Flores Anato, Joseph Cox, Daniel Grace, et al. "Characteristics of the sexual networks of gay, bisexual, and other men who have sex with men in Montreal, Toronto, and Vancouver: implications for the transmission and control of mpox". *Journal of Infectious Diseases (JID)* 229.S2 (Apr. 2024), pp. 293–304. doi.org/10.1093/infdis/jiae033.
7. Mackenzie Hamilton, **Jesse Knight**, and Sharmistha Mishra. "Examining the influence of imbalanced social contact matrices in epidemic models". *American Journal of Epidemiology (AJE)* 193.2 (Feb. 2024), pp. 339–347. doi.org/10.1093/aje/kwad185.
8. Korryn Bodner, **Jesse Knight**, Mackenzie Hamilton, and Sharmistha Mishra. "Testing whether higher contact among the vaccinated can be a mechanism for observed negative vaccine effectiveness". *American Journal of Epidemiology (AJE)* 192.8 (Mar. 2023), pp. 1335–1340. doi.org/10.1093/aje/kwad055.
9. **Jesse Knight**, Darrell H S Tan, and Sharmistha Mishra. "Maximizing the impact of limited vaccine supply under different early epidemic conditions: a 2-city modelling analysis of monkeypox virus transmission among men who have sex with men". *Canadian Medical Association Journal (CMAJ)* 194.46 (Nov. 2022), e1560–e1567. doi.org/10.1503/cmaj.221232.
10. **Jesse Knight**, Rupert Kaul, and Sharmistha Mishra. "Risk heterogeneity in compartmental HIV transmission models of ART as prevention in Sub-Saharan Africa: a scoping review". *Epidemics* 40 (Sept. 2022), p. 100608. doi.org/10.1016/j.epidem.2022.100608.
11. **Jesse Knight**, Huiting Ma, Amir Ghasemi, Mackenzie Hamilton, Kevin Brown, and Sharmistha Mishra. "Adaptive data-driven age and patch mixing in contact networks with recurrent mobility". *MethodsX* 9 (Jan. 2022), p. 101614. doi.org/10.1016/j.mex.2021.101614.
12. Sharmistha Mishra, Romain Silhol, **Jesse Knight**, Refilwe Phaswana-Mafuya, Daouda Diouf, Linwei Wang, Sheree Schwartz, Marie-Claude Boily, and Stefan D Baral. "Estimating the epidemic consequences of HIV prevention gaps among key populations". *Journal of the International AIDS Society (JIAS)* 24.S3 (July 2021), e25739. doi.org/10.1002/jia2.25739.
13. Linwei Wang, Nasheed Moqueet, Anna Simkin, **Jesse Knight**, et al. "Mathematical modelling of the influence of serosorting on the population-level HIV transmission impact of pre-exposure prophylaxis". *AIDS* 35.7 (June 2021), pp. 1113–1125. doi.org/10.1097/QAD.0000000000002826.
14. **Jesse Knight** and Sharmistha Mishra. "Estimating effective reproduction number using generation time versus serial interval, with application to Covid-19 in the Greater Toronto Area, Canada". *Infectious Disease Modelling (IDM)* 5 (Oct. 2020), pp. 889–896. doi.org/10.1016/j.idm.2020.10.009.
15. **Jesse Knight**, Stefan D Baral, Sheree Schwartz, Linwei Wang, Huiting Ma, Katherine Young, Harry Hausler, and Sharmistha Mishra. "Contribution of high risk groups' unmet needs may be underestimated in epidemic models without risk turnover: a mechanistic modelling analysis". *Infectious Disease Modelling (IDM)* 5 (July 2020), pp. 549–562. doi.org/10.1016/j.idm.2020.07.004.
16. Linwei Wang, Nasheed Moqueet, Gilles Lambert, Daniel Grace, et al. "Population-level sexual mixing according to HIV status and preexposure prophylaxis use among men who have sex with men in Montreal, Canada: implications for HIV prevention". *American Journal of Epidemiology (AJE)* 189.1 (Oct. 2019), pp. 44–54. doi.org/10.1093/aje/kwz231.
17. Hugo J Kuijff, J Matthijs Biesbroek, Jeroen de Bresser, Rutger Heinen, et al. "Standardized assessment of automatic segmentation of white matter hyperintensities and results of the WMH segmentation challenge". *IEEE Transactions on Medical Imaging (TMI)* 38.11 (Mar. 2019), pp. 2556–2568. doi.org/10.1109/TMI.2019.2905770.
18. **Jesse Knight**, Graham Taylor, and April Khademi. "Voxel-wise logistic regression and leave-one-source-out cross validation for white matter hyperintensity segmentation". *Magnetic Resonance Imaging (MRI)* 54 (Dec. 2018), pp. 119–136. doi.org/10.1016/j.mri.2018.06.009.
19. Olivier Commowick, Audrey Istace, Michael Kain, Baptiste Laurent, et al. "Objective evaluation of multiple sclerosis lesion segmentation using a data management and processing infrastructure". *Scientific Reports* 8.1 (Sept. 2018), p. 13650. doi.org/10.1038/s41598-018-31911-7.
20. **Jesse Knight**, Graham W Taylor, and April Khademi. "Equivalence of histogram equalization, histogram matching and the Nyul algorithm for intensity standardization in MRI". *Journal of Computational Vision and Imaging Systems* 3.1 (Oct. 2017). doi.org/10.15353/vsnl.v3i1.170.

21. Dan C Huynh, Mark W Parsons, Max Wintermark, Achala Vagal, et al. "Can CT perfusion accurately assess infarct core?" *Neurovascular Imaging* 2.1 (Mar. 2016), p. 7. doi.org/10.1186/s40809-016-0018-1.
22. Evgenia Maria Fanou, **Jesse Knight**, Richard I Aviv, Seyed-Parsa Hojjat, Sean P Symons, Lining Zhang, and Max Wintermark. "Effect of collaterals on clinical presentation, baseline imaging, complications, and outcome in acute stroke". *American Journal of Neuroradiology (AJNR)* 36.12 (Dec. 2015), pp. 2285–2291. doi.org/10.3174/ajnr.A4453.

ARTICLES IN PREPARATION

1. Domonique M Reed, Leigh F Johnson, Katherine Keyes, **Jesse Knight**, and Jeffrey W Imai-Eaton. "Hazardous alcohol use, sexual behavior, and incident HIV across 11 Eastern and Southern African countries". *AIDS* (revising). doi.org/10.64898/2026.03.30.26349734.
2. **Jesse Knight**, Siyi Wang, and Sharmistha Mishra. "Duration considerations for cross-sectional estimates of sexual behaviour: developing bias adjustments using Bayesian hierarchical models". *TBD* (preprint). doi.org/10.1101/2023.08.16.23294164.
3. **Jesse Knight** and Sharmistha Mishra. "Beyond instantaneous partnerships in compartmental models of HIV transmission: re-examining assumptions and a new incidence equation". *TBD* (drafting). github.com/mishra-lab/hiv-model-eswatini/tree/master/docs/foi.
4. Rebecca L Anderson, **Jesse Knight**, Yuetian Ma, Adelina Arteni, et al. "Systematic review on the prevalence of stigma in healthcare settings among sexual and gender minorities in Africa and the Middle East, 2010-2024". *TBD* (drafting).
5. **Jesse Knight**, Michael Pickles, and Marie-Claude Boily. "Mechanistic determinants of major depressive disorder current/lifetime prevalence: an individual-based modelling study". *TBD* (drafting).
6. **Jesse Knight**, Rebecca L Anderson, Romain Silhol, and Marie-Claude Boily. "What can prevalence-based measures of association tell us about hazard ratios? An individual-based modelling study of abuse and depression". *TBD* (drafting).

CONFERENCES (SELECTIVE)

1. Rebecca Anderson, Oli Stevens, **Jesse Knight**, Dennis Adjei Annang, et al. "The Key Population Data Initiative: Using individual participant data meta-analysis for granular estimates of HIV prevalence, age, and duration at-risk in sub-Saharan Africa". *Poster: International AIDS Conference (IAS)*. Vol. 26. July 2026. programme.aids2026.org/Abstract/Abstract/?abstractid=4317.
2. **Jesse Knight**, Michael Pickles, James Stannah, Jack Stone, Peter Vickerman, and Marie-Claude Boily. "Inferring incidence rates ratios (IRR) of hazardous drinking onset and recovery while depressed from cross-sectional adjusted odds ratios (AOR) using individual-based simulation modelling". *Oral: Canadian Society for Epidemiology and Biostatistics Conference (CSEB)*. Aug. 2025. cseb.ca/events/2025-conference/.
3. Fanyu Xiu, Jorge Luis Flores Anato, Joseph Cox, Daniel Grace, et al. "Characteristics of the sexual networks of gay, bisexual, and other men who have sex with men in Montréal, Toronto, and Vancouver: implications for the transmission and control of mpox in Canada". *Oral: International Conference on Infectious Disease Dynamics (Epidemics)*. Vol. 9. Nov. 2023. elsevier.com/events/conferences/all/international-conference-on-infectious-disease-dynamics.
4. Ruth Mwatelah, **Jesse Knight**, Huiting Ma, Kennedy Olango, et al. "Developing a screening tool to reach men who have sex with men living with undiagnosed HIV in Kenya". *Poster: Canadian Conference on HIV/AIDS Research (CAHR)*. Apr. 2023. www.cahr-acrv.ca/conference/.
5. **Jesse Knight** and Sharmistha Mishra. "Beyond instantaneous partnerships: capturing partnership-level herd effects in compartmental models of sexually transmitted infections". *Poster: Society for Medical Decision Making North American Meeting (SMDM)*. Vol. 44. Nov. 2022. www.smdmmeeting.org/.
6. Mackenzie Hamilton, **Jesse Knight**, and Sharmistha Mishra. "Failure to balance social contact matrices can bias models of SARS-CoV-2 transmission". *Poster: Society for Medical Decision Making North American Meeting (SMDM)*. Vol. 44. Nov. 2022. www.smdmmeeting.org/.

7. **Jesse Knight**, Huiting Ma, Amir Ghasemi, Mackenzie Hamilton, Kevin Brown, and Sharmistha Mishra. "Contact patterns by age and geography with recurrent mobility: influence of relaxing assumptions". *Poster: International Conference on Infectious Disease Dynamics (Epidemics)*. Vol. 8. Nov. 2021. elsevier.com/events/conferences/all/international-conference-on-infectious-disease-dynamics.
8. **Jesse Knight** and Sharmistha Mishra. "Importance of high risk groups may be further underestimated without simulated turnover in STI epidemics with assortative mixing". *Poster: Canadian Student Health Research Forum*. Aug. 2020. [umanitoba.ca/outreach/conferences/research`days/](https://umanitoba.ca/outreach/conferences/research%20days/).
9. Linwei Wang, Nasheed Moqueet, Anna Simkin, **Jesse Knight**, et al. "Influence of serosorting on the population-level HIV transmission impact of pre-exposure prophylaxis (PrEP) among men who have sex with men (MSM): a mathematical modeling study". *Poster: International AIDS Conference (IAS)*. Vol. 23. July 2020. programme.aidso2020.org/Abstract/Abstract/5283.
10. Linwei Wang, Nasheed Moqueet, Gilles Lambert, Daniel Grace, et al. "Quantifying sexual mixing by HIV status and Pre-Exposure Prophylaxis (PrEP) use among men who have sex with men". *Poster: STI & HIV 2019 World Congress (ISSTD)*. July 2019. doi.org/10.1136/sextrans-2019-sti.592.
11. **Jesse Knight**, Linwei Wang, Huiting Ma, Sheree Schwartz, Stefan Baral, and Sharmistha Mishra. "The influence of risk group turnover in STI/HIV epidemics: mechanistic insights from transmission modeling". *Oral: STI & HIV 2019 World Congress (ISSTD)*. July 2019. doi.org/10.1136/sextrans-2019-sti.214.
12. **Jesse Knight**, Huiting Ma, Stefan D Baral, Sosthenes Ketende, et al. "Who is left in 10-10-10? Importance of reaching key populations with the HIV cascade". *Poster: Conference on Retroviruses and Opportunistic Infections (CROI)*. Mar. 2019. croiconference.org/sessions/who-left-10-10-10.
13. **Jesse Knight**, Graham W Taylor, and April Khademi. "Voxel-Wise Logistic Regression for white matter hyperintensity segmentation in FLAIR MRI". *Competitor: WMH Segmentation Challenge at MICCAI (MICCAI)*. Sept. 2017. wmh.isi.uu.nl/#Toc122355715.
14. **Jesse Knight** and April Khademi. "MS lesion segmentation using FLAIR MRI only". *Competitor: MSSEG Challenge at MICCAI (MICCAI)*. Oct. 2016. www.hal.inserm.fr/inserm-01397806.
15. Seyed-Parsa Hojjat, **Jesse Knight**, Dan C Huynh, Daniel Eftekhari, Rita Vitorino, David Gladstone, Sean P Symons, and Richard I Aviv. "Topographic mapping of collateral impacts: effect of occlusion location and recanalization status in acute ischemic stroke". *Oral: Canadian Stroke Congress*. Sept. 2016. doi.org/10.1177/1747493016659793.
16. **Jesse Knight**, Alan R Moody, and April Khademi. "Noise in parallel MRI: how to determine whether single-coil assumptions still hold (they don't)". *Poster: Imaging Network Ontario Symposium (ImNO)*. Mar. 2016. doi.org/10.13140/RG.2.2.11028.91527.

* Competitor: trained model on challenge dataset; containerized model for blinded evaluation by organizers; drafted methods abstract; presented methods at challenge day; discussed challenge results at roundtable.

PRESENTATIONS (INVITED & NON-REVIEWED)

1. **Jesse Knight**. "Modelling the prevention impacts of cascade equality in Eswatini". *Seminar: HIV Modelling Consortium*. Feb. 2026.
2. **Jesse Knight**. "Estimating vaccine efficacy: the role of transmission modelling". *Invited Talk: 2025 Emerging & Pandemic Infections Consortium Symposium (EPIC)*. Oct. 2025. epic.utoronto.ca/event/2025-epic-symposium/.
3. **Jesse Knight**. "Beyond instantaneous partnerships: re-examining the force of infection equation in compartmental HIV transmission models". *Seminar: Colloquium on Mathematics for Public Health*. Feb. 2024.
4. **Jesse Knight**. "Adjusting for Biases in Sexual Behaviour Data". *Seminar: Dry Bench Seminar, Unity Health Toronto Research Training Center*. Nov. 2023.
5. **Jesse Knight** and Korryn Bodner. "Introduction to infectious disease modelling". *Guest Lecture: HAD5738 Advanced Methods for Economic Evaluation, University of Toronto*. June 2023.
6. **Jesse Knight**. "Beyond instantaneous partnerships: re-examining the force of infection equation in compartmental HIV transmission models". *Seminar: HIV Modelling Consortium*. May 2023.

7. Sharmistha Mishra, **Jesse Knight**, Trystan Leng, Romain Silhol, et al. "Methods for Key Populations Indicators". *Proposal: UNAIDS Reference Group on Estimates, Modelling, and Projections*. May 2023.
8. **Jesse Knight**, Linwei Wang, Oliver Gatalo, Huiting Ma, and Sharmistha Mishra. "Mpox network transmission modelling". *Meeting: Toronto Public Health*. Sept. 2022.
9. **Jesse Knight**. "Mixing by patch & age with recurrent mobility for Covid-19". *Seminar: Canadian Network for Modelling Infectious Diseases*. Sept. 2022.
10. **Jesse Knight**. "Compartmental modelling of infection transmission at scale applied to Eswatini". *Seminar: Thursday Seminar Series, Vector Institute*. July 2022.
11. Sharmistha Mishra, **Jesse Knight**, Oliver Gatalo, and Darrell HS Tan. "Mpox mathematical modeling". *Meeting: Ontario Mpox Community Mobilization*. June 2022.
12. **Jesse Knight** and Sharmistha Mishra. "Heterosexual HIV transmission in Eswatini: a descriptive modelling analysis". *Poster: IMS Scientific Day*. May 2022.
13. **Jesse Knight**. "Table 1 for transmission models: plots to describe simulated epidemics with multiple risk groups and contact types". *Seminar: Dry Bench Seminar, Unity Health Toronto Research Training Center*. Mar. 2022.
14. Korryn Bodner, **Jesse Knight**, Mackenzie Hamilton, and Sharmistha Mishra. "Introduction to infectious disease modelling". *Guest Lecture: HAD5738 Advanced Methods for Economic Evaluation, University of Toronto*. Feb. 2022.
15. **Jesse Knight**. "Mixing by Patch & age with recurrent mobility for Covid-19". *Seminar: MfPH Next Generation Seminar Series*. Nov. 2021.
16. **Jesse Knight** and Sharmistha Mishra. "Time between infections versus time between symptom onset in Covid-19: implications for estimating the reproduction number". *Seminar: Canadian Society of Applied and Industrial Mathematics Annual Meeting (CAIMS) Mini-Symposium: Mathematical modeling of Covid-19 transmission and mitigation strategies*. June 2021.
17. **Jesse Knight**. "White matter lesion segmentation: hyperparameter optimization for false positive reduction in FLAIR MRI". *Seminar: Guelph Engineering Society Graduate Engineering Symposium*. Mar. 2016.
18. **Jesse Knight**. "Engineering equipment design for the developing world". *Guest Lecture: ENGG 3100 Engineering Design III, University of Guelph*. Jan. 2016.
19. **Jesse Knight**. "Mapping collateral impacts: regional sparing in acute ischemic stroke". *Poster: Sunnybrook Research Institute Summer Research Project Day*. Aug. 2014.

BOOK CHAPTERS

1. **Jesse Knight** and Sharmistha Mishra. "Contact Matrices in Compartmental Disease Transmission Models". *Mathematics of Public Health*. Ed. by Jummy David and Jianhong Wu. Springer Nature, Dec. 2023, pp. 87–110. doi.org/10.1007/978-3-031-40805-2.
2. Brittany Reiche, **Jesse Knight**, Alan R Moody, and April Khademi. "Segmentation and Characterization of WML in FLAIR MRI". *Medical Image Analysis and Informatics: Computer-aided Diagnosis and Therapy*. Ed. by Paulo Mazzoncini de Azevedo Marques, Arianna Mencattini, Marcello Salmeri, and Rangaraj M Rangayyan. Vol. 2. CRC Press, Nov. 2017, pp. 1–28. doi.org/10.1201/9781351228343.
3. **Jesse Knight** and April Khademi. "Disease-Inspired Feature Design for Computer-Aided Diagnosis of Breast Cancer Digital Pathology Images". *Medical Image Analysis and Informatics: Computer-aided Diagnosis and Therapy*. Ed. by Paulo Mazzoncini de Azevedo Marques, Arianna Mencattini, Marcello Salmeri, and Rangaraj M Rangayyan. Vol. 2. CRC Press, Nov. 2017, pp. 427–450. doi.org/10.1201/9781351228343.

THESES

1. **Jesse Knight**. "Reexamining Assumptions in Compartmental Models of Heterosexual HIV Transmission applied to Eswatini". Doctor of Philosophy. University of Toronto, Nov. 2023. hdl.handle.net/1807/129897.
2. **Jesse Knight**. "Voxel-Wise Image Analysis for White Matter Hyperintensity Segmentation". Master of Applied Science. University of Guelph, Apr. 2018. hdl.handle.net/10214/12142.

PEER REVIEW: JOURNAL ARTICLES

2026.03 – present (1) [BMC Infectious Diseases](#)
2025.11 – present (3) [PLOS Computational Biology](#)
2025.06 – present (1) [Vaccine](#)
2025.03 – present (3) [Bulletin of Mathematical Biology](#)
2025.03 – present (2) [Computers in Biology and Medicine](#)
2025.01 – present (3) [Nature Communications](#)
2024.07 – present (1) [American Journal of Epidemiology](#)
2024.05 – present (3) [PLOS Global Public Health](#)
2024.05 – present (1) [Journal of Acquired Immune Deficiency Syndromes](#)
2024.03 – present (4) [PLOS One](#)
2024.03 – present (2) [BMC Medicine](#)
2024.01 – present (5) [Scientific Reports](#)
2022.10 – present (5) [Epidemiology & Infection](#)
2021.04 – present (3) [Annals of Epidemiology](#)
2020.08 – 2023.05 (5) [Canadian Medical Association Journal Open](#)
2018.05 – 2021.03 (9) [Journal of Digital Imaging](#)
2015.11 – 2016.09 (2) [Canadian Journal of Electrical and Computer Engineering](#)

PEER REVIEW: CONFERENCE ABSTRACTS

2024.06 – 2026.02 (40) [Society for Medical Decision Making Annual Meeting](#)
2017.04 (1) [IEEE International Humanitarian Technology Conference](#)
2016.02 (2) [Canadian Conference of Electrical and Computer Engineering](#)

Media & Science Communication

ARTICLES (INTERVIEWED)

2022.12 [New Model Aims to Aid Monkeypox Vaccine Distribution](#), Medscape
2022.09 [Toronto Monkeypox Studies Suggest Ways to Reduce Viral Spread](#), U of T Medicine Dean's Report
2022.09 [Science Literacy Week 2022: Meet two mathematical modellers](#), U of T Emerging & Pandemic Infections Consortium
2022.01 [Raw Talk Podcast Celebrates 100th Episode with Insulin 100](#), IMS Magazine
2020.12 [Thanks to Alumni Support, Raw Talk Podcast Tells Powerful Stories About The Science of Medicine](#), U of T Alumni Website

PODCASTS

Guest

2023.04 [Maximizing the impact of limited vaccine supply under different early mpox epidemic conditions](#), Mod4PH Research Highlights

Producer & Show Host, Raw Talk Podcast

2022.02 [#104 A New Era of Knowledge: Science on Social Media](#)
2021.10 [#100 Years Later: Insulin and Beyond](#)
2021.09 [#98 Podium Pills: Fame or Folly?](#)
2021.09 [#97 Let's Talk Grad School](#)

2021.05 #96 Genetic Testing: What it Means for Families and the Future
 2021.04 #94 Child Maltreatment: Healing and Breaking the Cycle
 2021.03 #92 Producing and Pricing Drugs in Canada
 2021.02 #90 Anti-Black Racism in Healthcare
 2021.01 #88 Tuberculosis: The Forgotten Pandemic
 2020.12 #86 Global Health
 2020.11 #84 Passports and Procedures: Receiving and Providing Care Abroad
 2020.10 #82 Making Strides: Amputation and Prosthetics
 2020.09 #80 COVID Decoded: Science and Society in the New Normal
 2020.02 #74 Outbreak Transmission: When Diseases Go Viral
 2020.01 #72 Burning Up: The Health Impacts of Climate Change

EVENTS

2021.06 **Our Health on Thin Ice, Tech & Content Co-Lead**, Raw Talk Podcast & U of T Talks
Two panel discussions on climate change & health
 2021.02 **Podcast Festival, Tech & Content Co-Lead**, Raw Talk Podcast
Virtual 2-day conference on podcasting & science communication
 2020.06 **COVID Decoded, Co-Host & Content**, Raw Talk Podcast
Live-streamed 8-part 1-hour YouTube series on COVID-19 topics

Skills & Research Interests

SKILLS

- **Code, Advanced:** R: base, ggplot, Stan; Python: Numpy, SciPy, Matplotlib, Pandas; $\text{\LaTeX}$
- **Code, Intermediate:** Matlab; bash; Git; web: Bootstrap, Jinja, jQuery
- **Analysis:** simulation modelling, epidemiology, biostatistics, Bayesian methods, systematic review

RESEARCH INTERESTS

- **Mathematical Modelling:** influence of model structure & assumptions on outputs; sensitivity & uncertainty analysis; new models of epidemiological phenomena; simulation-based study design validation
- **Infectious Disease Modelling:** risk heterogeneity & contact patterns; highly stratified compartmental models; individual-based models; dynamic network-based models
- **HIV & STI Modelling:** complex sexual networks; men who have sex with men; sex work; sexual life course; incorporating structural factors; intervention impact estimation; qualitative insights; combating stigma
- **Data Science:** intuitive visualization; bias identification & adjustment; Bayesian methods; causal mediation analysis; optimization; parallel computing; reusable & open-source code